UNDERSTANDING

MATHEMATICAL INDUCTION

Prove It for All n!

WHAT DOES "FOR ALL N" MEAN?

"For all n" means for every number starting from 1, 2, 3, and so on forever! Think of it like this: if you can jump 1 step, then 2 steps, then 3 steps... can you jump ALL the steps?

It includes every natural number, no matter how big!

We prove something works for 1, 2, 3, 4, 5... forever!

WHY DO WE NEED TO PROVE FOR ALL N?

Sometimes we want to show things are true forever, not just for one number. Mathematical induction helps us prove patterns work for every single number!

Helps understand patterns in mathematics

Builds strong math and reasoning skills

What Is Mathematical Induction?

A powerful way to prove things

Works for infinite numbers!

Step 1: Base case (first step)

Step 2: Inductive step (next step)

Proves truth for all n forever!

Mathematical induction is like climbing stairs: if you can get on the first step, and from any step you can get to the next, you can climb all stairs!

STEP 1 — THE BASE CASE

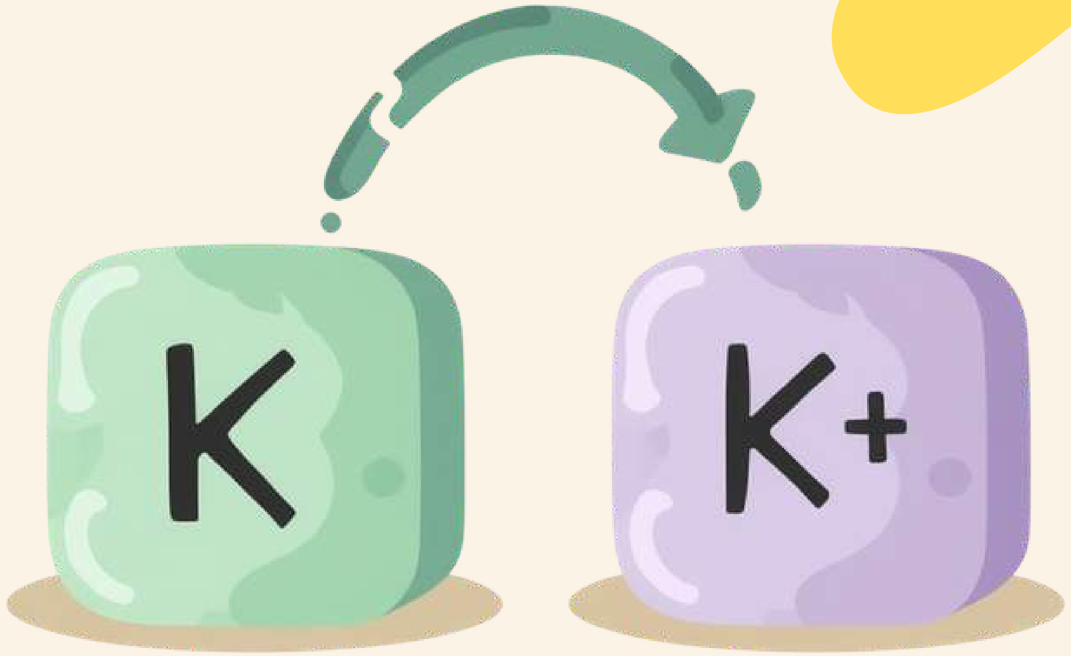
First, prove the statement is true for the first number, usually $n=1$. This is like making sure you can step onto the first stair before climbing higher!

Example: Check if $1 = 1(1+1)/2 = 1$ ✓

The base case is your starting point for proof

STEP 2 — THE INDUCTIVE STEP

Next, assume the statement is true for n = k, and then prove it for n = k + 1. This is the magic step that connects everything together!



Assume true for k (called the induction hypothesis)

Show it works for k+1 using your assumption

Putting It Together: Full Induction Proof

- Prove base case (n=1)
- Assume true for n=k
- Prove true for n=k+1 using assumption
- Conclude true for all n by induction
- Done! It works forever!

Follow these four steps to complete any mathematical induction proof and prove your statement is true for all natural numbers!

Example 1: Sum of First n Natural Numbers

Base Case: Check $n = 1$

Left side: 1, Right side: $1(1+1)/2 = 1$ ✓

Inductive Step: Assume true for $n = k$

Then $1+2+\ldots+k = k(k+1)/2$

Prove for $k+1$: Add $(k+1)$ to both sides!

Statement: $1 + 2 + 3 + \ldots + n = n(n+1)/2$. Let's prove this is true for ALL natural numbers using mathematical induction!

Example 2: Multiples of 3

Base Case (n=1):
$3 \times 1 = 3$
Is 3 a multiple of 3? Yes!
$3 \div 3 = 1$ (no remainder)
✓ Base case is TRUE!

Inductive Step: From k to k+1

Assume 3k is a multiple of 3.
Now check $3(k+1) = 3k + 3$
Both 3k and 3 are multiples of 3,
so their sum is also a multiple of 3!
✓ Inductive step is TRUE!

Powers of 2 Base Case Check

Prove: $2^n \geq n + 1$ for all $n \geq 1$
Base Case ($n = 1$):
- Left side: $2^1 = 2$
- Right side: $1 + 1 = 2$
- Since $2 \geq 2$ is true ✓

The base case works!

The Inductive Step Explained

Inductive Step:
- Assume $2^k \geq k + 1$ is true for some k
- Prove $2^{k+1} \geq (k + 1) + 1$
- Since $2^{k+1} = 2 \times 2^k \geq 2(k + 1)$
- And $2(k + 1) \geq k + 2$ when $k \geq 1$ ✓

Common Mistakes to Avoid

- Forgetting to prove the base case (n=1)
- Not clearly stating the induction hypothesis
- Skipping the step from k to k+1
- Assuming what you need to prove
- Forgetting to write the conclusion

WHY DOES INDUCTION WORK?

"Induction is a magical ladder to climb infinitely many steps!"

Induction works because once we prove the first step is true, and we show that any step leads to the next, we create an unbreakable chain of truth that goes on forever!

Recap — What We Learned Today

"For all n" means proving for every number 1, 2, 3...

Base case: Always start by proving n=1 works

Inductive step: Show k → k+1 connection

Full proof: Base case + Inductive step = True for all n!

You can now climb the infinite staircase of math!

Try It Yourself! Practice Time!

Challenge 1: Sum of Odd Numbers
Prove using induction:
$1 + 3 + 5 + ... + (2n-1) = n^2$
Hint: Check n=1 first, then assume true for k and prove for k+1!

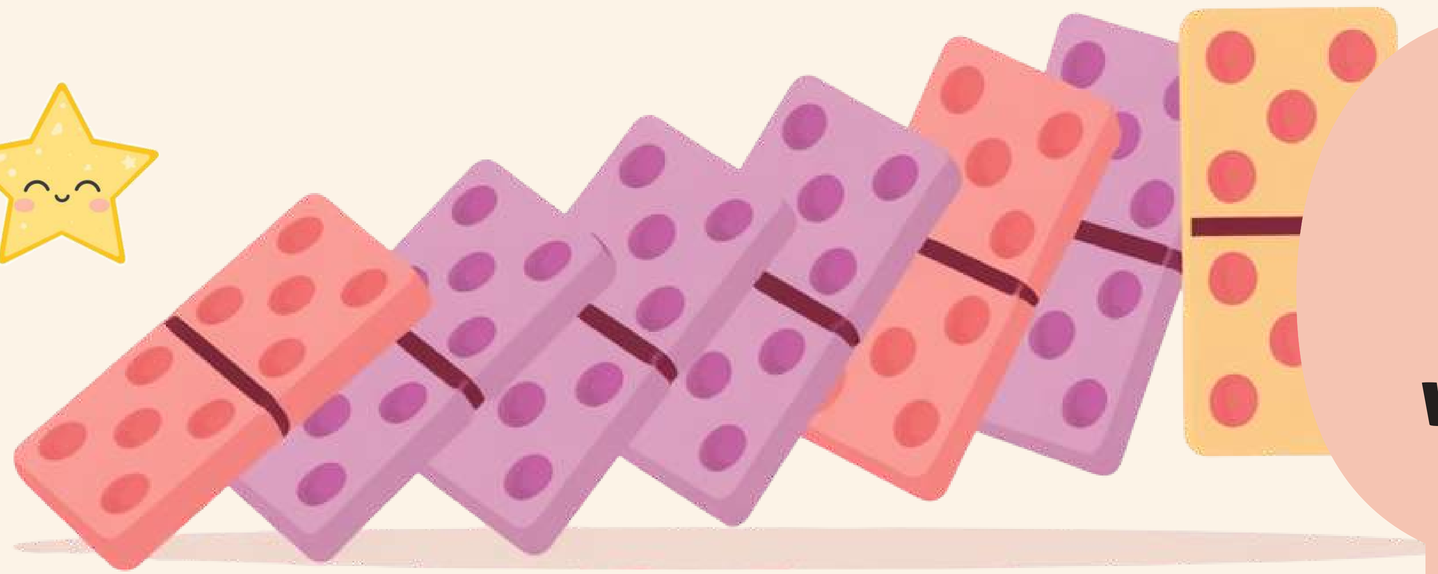
Challenge 2: Powers of 5

Prove using induction:
5^n is divisible by 5 for all $n \geq 1$
Hint: What is 5^1? Can you show 5^(k+1) is also divisible by 5?

QUESTIONS & THANK YOU!

Thanks for learning Mathematical Induction with us today! We hope you enjoyed climbing the infinite staircase of proofs. If you have any questions about base cases, inductive steps, or anything else, now is the time to ask. Keep practicing and you'll master proving things "for all n" in no time!

BONUS: FUN INDUCTION PUZZLE



If the first domino falls and every domino knocks over the next, what happens to all the dominoes?

This is induction in real life! Just like proving math for all n, dominoes show that one step leads to the next, forever!

Think About It!